

Logic Gates — Mark schemes

PAPERS A TO D • 30 MARKS EACH • TUTOR COPY

Marks in square brackets show how the total is split, one per line. Marks on this topic are lost to a short list of habits: ignoring the small circle on NAND and NOR, reading OR as “one or the other”, attaching NOT to the wrong thing (NOT A AND B against NOT (A AND B)), missing a row in a truth table, and testing rows that cannot tell two circuits apart. Every note below names the mistake that question was built to catch and shows the wrong column it produces, so that you can recognise the pattern when it turns up in a different paper. Every table here was computed by enumerating all the input combinations.

Paper A — Medium

30 MARKS

- 1** (a) **8 rows ($2 \times 2 \times 2$) [1]** [3]
 (b) Eight different rows, none missing and none repeated [1]
 In counting order: 000, 001, 010, 011, 100, 101, 110, 111 [1]
The addition slip gives $2 + 2 + 2 = 6$ rows; multiplying gives 8. Writing the rows in the order a person happens to think of them (for example by how many 1s each has: 000, 001, 010, 100, 011, 101, 110, 111) often keeps all eight but loses the order mark: award the first mark of (b) only.
- 2** AND column: **0, 0, 0, 1 [1]** [2]
 OR column: **0, 1, 1, 1 [1]**
The OR trap: reading OR as “one or the other” writes 0 on the 1 1 row and gives 0, 1, 1, 0. That is XOR, which this topic does not examine. Award the AND mark and withhold the OR mark.
- 3** NAND column: **1, 1, 1, 0 [1]** [2]
 NOR column: **1, 0, 0, 0 [1]**
Ignoring the circle copies the plain gate: NAND written as 0, 0, 0, 1 (AND's column) and NOR written as 0, 1, 1, 1 (OR's column). Each mark is independent.
- 4** (i) **NOR [1]** [4]
 (ii) **NOT [1]**
 (iii) **AND [1]**
 (iv) **NAND [1]**
Only one of the four symbols has a single input: the NOT. A circle on its own is not a gate, so calling (ii) NAND or NOR is the “circle means NAND” trap. For the other three, a curved back is the OR family (NOR) and a flat back is the AND family (AND, NAND); then the circle decides between AND and NAND.

- 5 (a) **$X = (A \text{ OR } B) \text{ AND } (\text{NOT } C)$** [1] (accept P AND R; brackets round NOT C are optional) [4]

(b) $P = A \text{ OR } B$: **0, 0, 1, 1, 1, 1, 1, 1** [1]

$R = \text{NOT } C$: **1, 0, 1, 0, 1, 0, 1, 0** [1]

$X = P \text{ AND } R$: **0, 0, 1, 0, 1, 0, 1, 0** [1] (award if consistent with their own P and R)

A	B	C	P	R	X
0	0	0	0	1	0
0	0	1	0	0	0
0	1	0	1	1	1
0	1	1	1	0	0
1	0	0	1	1	1
1	0	1	1	0	0
1	1	0	1	1	1
1	1	1	1	0	0

Reading the circuit without its brackets, as $A \text{ OR } (B \text{ AND } \text{NOT } C)$, gives 0, 0, 1, 0, 1, 1, 1, 1. Dropping the NOT on C ($X = P \text{ AND } C$) gives 0, 0, 0, 1, 0, 1, 0, 1. Both are wrong in several rows against the correct 0, 0, 1, 0, 1, 0, 1, 0, which has 3 ones. The method on trial is filling one column at a time.

- 6 (a) **$X = (\text{NOT } A) \text{ OR } B$** : **1, 1, 0, 1** [1] [3]

$Y = \text{NOT } (A \text{ OR } B)$: **1, 0, 0, 0** [1]

(b) **NOR**, and it is **Y** [1]

Reading X as $\text{NOT } (A \text{ OR } B)$ gives Y 's column 1, 0, 0, 0: the NOT was applied to the whole expression instead of to A alone. X matches no single gate: it has three 1s like OR and NAND, but its 0 is on the row 10, whereas OR's is on 00 and NAND's on 11.

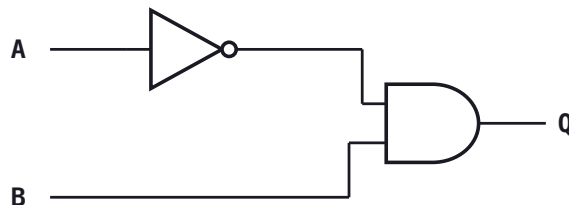
- 7 (a) **$Q = (\text{NOT } A) \text{ AND } B$** [1] (NOT A AND B is accepted under this course's rule) [5]

(b) $Q = 0, 1, 0, 0$: the row $A = 0, B = 1$ gives 1 [1]

the other three rows give 0 [1]

(c) A NOT gate on A [1]

An AND gate taking NOT A and B, with its output labelled Q [1]



Reading the sentence as $\text{NOT } (A \text{ AND } B)$ gives 1, 1, 1, 0: three 1s instead of one. A drawing that uses a NAND or a NOR scores nothing in (c), and a NOT drawn with two inputs loses the first mark. Mark the structure; do not penalise untidy symbols.

- 8 (a) **$H = (\text{NOT } M) \text{ AND } (\text{NOT } P)$** [1] [4]

(b) The row $M = 0, P = 0$ gives 1 [1]

The other three rows give 0, so $H = 1, 0, 0, 0$ [1]

(c) **NOR** [1]

$\text{NOT } (M \text{ AND } P)$ and $(\text{NOT } M) \text{ OR } (\text{NOT } P)$ both give 1, 1, 1, 0, which is NAND's column: they say "at least one is off", but the lamp needs both off. The sentence has "and" in it, and each condition needs its own NOT.

- 9 (a) $A = 0$ gives **1** [1] [3]

$A = 1$ gives **0** [1]

(b) **NOT** [1]

Forgetting the circle writes $Z = 0, 1$: AND of A with itself is just A. The joined inputs make NOT $(A \text{ AND } A)$, and $A \text{ AND } A$ is A, so the circle leaves the opposite of A.

- 1 AND: **0, 0, 0, 0, 0, 0, 0, 1** [1] [2]
OR: **0, 1, 1, 1, 1, 1, 1, 1** [1]

Reading OR as “exactly one input is 1” gives 0, 1, 1, 0, 1, 0, 0, 0: it drops the rows with two or three 1s, which OR counts too. Award the AND mark only.

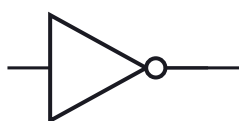
- 2 (a) **16** [1] [3]
(b) **64** [1]
(c) **5** ($2^5 = 32$) [1]

Rows double with every extra input: 2, 4, 8, 16, 32, 64. The slip is to add: 4 inputs by addition is $2 + 2 + 2 + 2 = 8$, or $4 \times 2 = 8$, and 6 inputs by the same habit gives 12. (c) works backwards: 32 is five doublings.

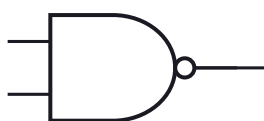
- 3 (i) **NAND** (0 only on the 1 1 row) [1] [4]
(ii) **AND** (1 only on the 1 1 row) [1]
(iii) **NOR** (1 only on the 0 0 row) [1]
(iv) **OR** (0 only on the 0 0 row) [1]

There are two confusable pairs. NAND (1, 1, 1, 0) and OR (0, 1, 1, 1) each have three 1s, but NAND's 0 is on row 11 and OR's on row 00. AND (0, 0, 0, 1) and NOR (1, 0, 0, 0) each have a single 1, on row 11 and row 00. A student who counts the 1s and stops cannot separate either pair.

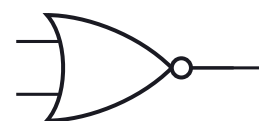
- 4 (a) **NOT**: a triangle with a small circle on its point; **one** input and one output [1] [3]
(b) **NAND**: the flat-backed D-shape of AND with a small circle on the output; two inputs [1]
(c) **NOR**: the curved-backed OR shape with a small circle on the output; two inputs [1]



(a)



(b)



(c)

Withhold the mark if the circle is missing (that draws AND, OR or half a NOT) or if the NOT has two inputs. Do not penalise a wobbly curve: look for three features only, which are the body shape, the circle and the number of inputs.

- 5 (a) **Q = (A OR B) NAND C** [1] (accept P NAND C) [5]
(b) P = A OR B: **0, 0, 1, 1, 1, 1, 1, 1** [1]
Q is 0 only where P = 1 and C = 1, which is rows 011, 101, 111 [1]
every other row gives 1, so Q = **1, 1, 1, 0, 1, 0, 1, 0** [1]
(c) **5** rows [1]

A	B	C	P	Q
0	0	0	0	1
0	0	1	0	1
0	1	0	1	1
0	1	1	1	0
1	0	0	1	1
1	0	1	1	0
1	1	0	1	1
1	1	1	1	0

Treating the NAND as an AND gives 0, 0, 0, 1, 0, 1, 0, 1, which is the exact opposite of the correct column in every row: the circle was ignored. (c) is follow-through: accept the number of 1s in the student's own Q column if it is the right count for that column.

- 6** (a) **$X = \text{NOT } (A \text{ AND } B)$ [1]** (the brackets are essential) [3]
 (b) **NAND [1]**
 (c) **$Y = A \text{ AND } (\text{NOT } B)$ [1]** (accept $A \text{ AND NOT } B$)

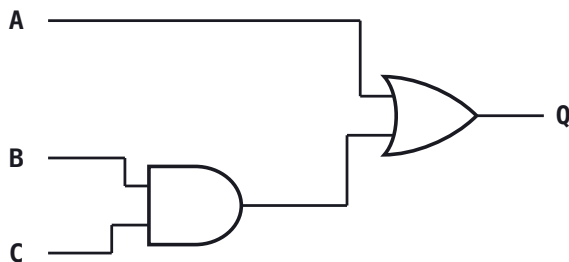
In circuit (i) the NOT comes after the AND, so the brackets must go round the AND. Writing it as NOT A AND B means (NOT A) AND B, whose column is 0, 1, 0, 0 against NAND's 1, 1, 1, 0. In circuit (ii) the NOT sits on B alone, so brackets round the whole AND would be wrong there.

- 7** (a) "Train approaching and barrier not down" is **$A \text{ AND } (\text{NOT } B)$ [1]** [5]
 Joined to the test switch by OR, with brackets round the AND part: **$Q = (A \text{ AND } (\text{NOT } B)) \text{ OR } C$ [1]**
 (b) The rows with $C = 1$ (001, 011, 101, 111) all give 1 [1]
 Row 100 gives 1 (train approaching, barrier up, test switch off) [1]
 Rows 000, 010 and 110 give 0, so $Q = 0, 1, 0, 1, 1, 1, 0, 1$ [1]

A	B	C	Q
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	1

Dropping the NOT (reading "not fully down" as plain B) gives 0, 1, 0, 1, 0, 1, 1, 1: row 100 turns to 0 and row 110 turns to 1. Joining with AND instead of OR gives 0, 0, 0, 0, 0, 1, 0, 0. Follow-through is allowed in (b) if the table matches the student's own expression.

- 8** (a) "Card slot blocked and camera covered" is **$B \text{ AND } C$ [1]** [5]
 Joined to A by OR, brackets round the AND: **$Q = A \text{ OR } (B \text{ AND } C)$ [1]**
 (b) An AND gate with inputs B and C [1]
 An OR gate taking A and the output of the AND [1]
 Every input labelled, output labelled Q, wires joined correctly [1]



The common bracket slip is $(A \text{ OR } B) \text{ AND } C$, whose column is 0, 0, 0, 1, 0, 1, 0, 1 against the correct 0, 0, 0, 1, 1, 1, 1, 1: they differ on rows 100 and 110. A drawing with the AND as the last gate is the wrong circuit even if each symbol is perfect. If the expression in (a) is wrong but the drawing in (b) matches it, award (b) in full.

- 1** (i) **NAND** and (iv) **OR**: both have seven 1s; the 0 of NAND is on 111, the 0 of OR on 000 [2]
[1]
(ii) **AND** and (iii) **NOR**: both have a single 1; the 1 of AND is on 111, the 1 of NOR on 000 **[1]**

Award a pair only if both names in it are right. The pairs are the confusable ones: the gates are told apart by where the odd digit sits, not by how many 1s there are. Naming (i) as OR and (iv) as NAND swaps the pair and scores nothing for it.

- 2** (a) **6** inputs ($2^6 = 64$) **[1]** [3]
(b) **8** rows (half of the 16) **[1]**
(c) **15** **[1]**

(b) Four is the tempting wrong answer: the number of inputs rather than the number of rows. (c) A NAND is 0 only on the row where AND is 1, which is 1111, so 15 of the 16 rows are 1. The answer 1 comes from copying AND's single 1 and ignoring the circle.

- 3** (a) **X = 0, 0, 0, 1, 0, 0, 0, 0** **[1]** [4]
Y = 1, 1, 1, 1, 1, 1, 1, 0 **[1]**
(b) **6** rows (000, 001, 010, 100, 101, 110): X is 0 and Y is 1 on each; on 011 and 111 they agree **[1]**
(c) **NAND** (three-input) **[1]**

A	B	C	X	Y
0	0	0	0	1
0	0	1	0	1
0	1	0	0	1
0	1	1	1	1
1	0	0	0	1
1	0	1	0	1
1	1	0	0	1
1	1	1	0	0

Reading X as NOT (A AND B AND C) makes it identical to Y, 1, 1, 1, 1, 1, 1, 0: seven 1s instead of one. NOT sticks to the thing straight after it, so in X the NOT belongs to A alone and the three conditions are "A off, B on, C on". A student who has X wrong will get (b) wrong as well, and (b) should be marked as follow-through: 0 differences if $X = Y$.

- 4** (a) **Q = (A AND B) OR (B NOR C)**: the two bracketed gates **[1]** [5]
joined by an OR, with brackets round each (accept P OR R) **[1]**
(b) **P = A AND B: 0, 0, 0, 0, 0, 0, 1, 1** **[1]**
R = B NOR C: 1, 0, 0, 0, 1, 0, 0, 0 **[1]**
Q = P OR R: 1, 0, 0, 0, 1, 0, 1, 1 **[1]**

A	B	C	P	R	Q
0	0	0	0	1	1
0	0	1	0	0	0
0	1	0	0	0	0
0	1	1	0	0	0
1	0	0	0	1	1
1	0	1	0	0	0
1	1	0	1	0	1
1	1	1	1	0	1

Treating the NOR as an OR ($R = B \text{ OR } C$) gives $Q = 0, 1, 1, 1, 0, 1, 1, 1$: the circle was ignored. B feeds both first-level gates, so a student who draws only one B wire into the picture in their head reads the circuit wrongly. Sanity check: the correct Q has 4 ones.

5 (a) Circuit **(ii)** [1] [3]

(b) Rules out (i): row **011 or 101**, where (i) gives 0 and the table says 1 [1]

Rules out (iii): row **001 or 111** (on 001 (iii) gives 0 against the table's 1; on 111 it gives 1 against 0) [1]

A	B	C	X	(i)	(ii)	(iii)
0	0	0	0	0	0	0
0	0	1	1	1	1	0
0	1	0	0	0	0	0
0	1	1	1	0	1	1
1	0	0	0	0	0	0
1	0	1	1	0	1	1
1	1	0	0	0	0	0
1	1	1	0	0	0	1

The decoys were built to agree with the table on six of the eight rows, so most rows prove nothing: all three circuits give the same answer on rows 000, 010, 100, 110. A student who checks those rows first can tick any circuit. The method is to test a row where the candidates disagree. A row quoted without the circuit's output on it is half an answer: award (b)'s marks only if the wrong output is stated.

6 (a) Rows **110 and 111** [1] [3]

(b) For 1 1 he gave 0: he treated OR as “one or the other but not both”, when OR gives 1 if at least one input is 1 [1]

(c) **1** cell (the Q in row 110) [1]

A	B	C	P	R	Q
0	0	0	0	1	1
0	0	1	0	0	1
0	1	0	1	1	0
0	1	1	1	0	1
1	0	0	1	1	0
1	0	1	1	0	1
1	1	0	1	1	0
1	1	1	1	0	1

His P column 0, 0, 1, 1, 1, 1, 0, 0 is XOR's, not OR's 0, 0, 1, 1, 1, 1, 1, 1. Two wrong cells in P produce only one wrong cell in Q: on row 111 the NOT gives R = 0, and a NAND with a 0 on either input outputs 1 whatever P is, so the error is hidden there. Accept 1 only for (c); the answer 2 counts the P cells.

7 (a) **A AND (NOT B)** for “too hot and the soil is not wet” [1] [5]

OR with C, then AND with D, brackets showing the order: **Q = ((A AND (NOT B)) OR C) AND D** [1]

(b) **5** rows [1]

D = 0 gives 0 on all 8 of its rows. With D = 1 the inner part is 1 when C = 1 (4 rows) or when A = 1, B = 0, C = 0 (1 row): 4 + 1 = 5 [1]

(c) Q = **0**: with A = 1 and B = 1 the first part is 0 (the soil is wet), and C = 0, so the OR is 0 and the AND with D is 0 [1]

Forgetting the pump (ignoring D) gives 10 rows: the inner part is 1 on 5 of its 8 rows, and doubling over both values of D gives 10. Breaking the bracket, (A AND NOT B) OR (C AND D), gives 7 rows. Neither is the sentence in the question, whose last clause makes D a veto.

- 8 (a) $P = 1, 1, 0, 0$ and $R = 1, 0, 1, 0$ [1]
 $Q = 0, 1, 1, 1$ [1]
 (b) A NAND with both inputs joined gives NOT (A AND A), and A AND A is just A, so the output is NOT A: 0 gives 1 and 1 gives 0 [1]
 (c) One NAND with both inputs wired to A and its output labelled Q [1]
 (d) OR [1]

A	B	P	R	Q
0	0	1	1	0
0	1	1	0	1
1	0	0	1	1
1	1	0	0	1



A student who thinks a joined NAND “copies” its input writes $P = 0, 0, 1, 1$ and $R = 0, 1, 0, 1$, and then Q (a NAND of those) = $1, 1, 1, 0$: that is NAND's own column, not OR's. If Q comes out as $1, 1, 1, 0$ the flip in the first two gates was missed.

Paper D — Hard

30 MARKS

- 1 $P = A \text{ NOR } B: 1, 1, 0, 0, 0, 0, 0, 0$ [1]
 $R = \text{NOT } C: 1, 0, 1, 0, 1, 0, 1, 0$ [1]
 $Q = P \text{ NOR } R: 0, 0, 0, 1, 0, 1, 0, 1$ [1]

A	B	C	P	R	Q
0	0	0	1	1	0
0	0	1	1	0	0
0	1	0	0	1	0
0	1	1	0	0	1
1	0	0	0	1	0
1	0	1	0	0	1
1	1	0	0	1	0
1	1	1	0	0	1

Q is 1 only when $P = 0$ and $R = 0$: that is when A or B is 1 and C is 1, rows 011, 101, 111. Using C instead of NOT C in the last gate gives 0, 0, 1, 0, 1, 0, 1, 0. If P and R are right but Q is wrong, the final NOR was applied as an OR: award the marks for P and R .

- 2 (a) 101 [1]
 (b) Row 7 [1]
 (c) 1001 [1]

Counting from 0 instead of 1 gives 110 for (a) (that is the seventh row), 6 for (b), and 1010 for (c). The first row is 000 (or 0000), so row n is the binary number for $n - 1$. The question states that rows are numbered from 1, so the off-by-one answers are marked wrong.

- 3 (a) OR [1]
 NAND [1]
 (b) 00 or 11: OR and NAND disagree on this row (on 00 OR gives 0 and NAND gives 1; on 11 OR gives 1 and NAND gives 0) [1]

Rows 01 and 10 give the same answer for both gates, so they tell you nothing: a student who offers one of them has not understood what “tell apart” asks. If a student names AND or NOR they have counted the 1s as one rather than three.

- 4 (a) $Q = (A \text{ NAND } B) \text{ NAND } (B \text{ NAND } C)$: the two inner NANDs [1] [5]
 joined by a third NAND, with brackets round each inner gate (accept P NAND R) [1]

(b) $P = A \text{ NAND } B$: 1, 1, 1, 1, 1, 1, 0, 0 [1]

$R = B \text{ NAND } C$: 1, 1, 1, 0, 1, 1, 1, 0 [1]

$Q = P \text{ NAND } R$: 0, 0, 0, 1, 0, 0, 1, 1 [1]

A	B	C	P	R	Q
0	0	0	1	1	0
0	0	1	1	1	0
0	1	0	1	1	0
0	1	1	1	0	1
1	0	0	1	1	0
1	0	1	1	1	0
1	1	0	0	1	1
1	1	1	0	0	1

Reading the NANDs as ANDs gives $Q = 0, 0, 0, 0, 0, 0, 0, 1$. In fact Q is 1 where $B = 1$ and at least one of A, C is 1 (rows 011, 110, 111). The last gate is the one to watch: P and R are both 1 on most rows, and a NAND of two 1s is 0.

- 5 (a) $X = 1, 1, 1, 0$ [1] [4]

$Y = 1, 1, 1, 0$ [1]

$Z = 1, 0, 0, 0$ [1]

(b) X and Y are identical, and both equal NAND [1]

A	B	X	Y	Z
0	0	1	1	1
0	1	1	1	0
1	0	1	1	0
1	1	0	0	0

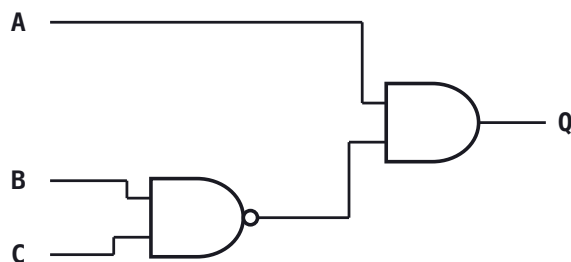
Z is NOR's column. A student who believes $\text{NOT}(A \text{ AND } B)$ is the same as $(\text{NOT } A) \text{ AND } (\text{NOT } B)$, with the NOT simply going inside, pairs X with Z . It does not: the NOT also changes the AND into an OR. That belief gives 1, 0, 0, 0 for X , against the true 1, 1, 1, 0.

- 6 (a) "Unless both are on" is $\text{NOT}(B \text{ AND } C)$, which is $B \text{ NAND } C$ [1] [4]

ANDed with the timer: $Q = A \text{ AND } (B \text{ NAND } C)$ (accept $A \text{ AND } (\text{NOT}(B \text{ AND } C))$) [1]

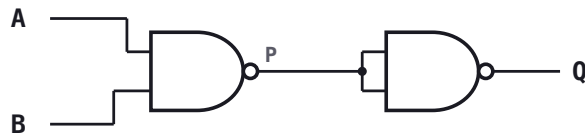
(b) A NAND gate taking B and C [1]

Its output and A going into an AND gate, with the output labelled Q (accept an AND, then a NOT, then an AND as the equivalent circuit) [1]



Reading "unless both" as "unless either" gives $A \text{ AND } (B \text{ NOR } C) = 0, 0, 0, 0, 1, 0, 0, 0$: the bell then falls silent whenever one switch is on. The question supplies the test row: only B on, timer firing must ring, which is row 110, where the right expression gives 1 and the wrong one gives 0.

- 7 (a) A first NAND taking A and B, its output labelled P **[1]** [3]
 A second NAND with both its inputs joined to P, output Q **[1]**
 (b) P = **1, 1, 1, 0** and Q = **0, 0, 0, 1** (the table for the drawn circuit; Q must equal A AND B) **[1]**



A	B	P	Q
0	0	1	0
0	1	1	0
1	0	1	0
1	1	0	1

If the second gate is read as passing P straight through, $Q = P = 1, 1, 1, 0$, which is NAND's column: the student has rebuilt NAND, not AND. The second gate is the joined-inputs NOT, so $Q = \text{NOT } P = 0, 0, 0, 1$.

- 8 (i) The mistake: the circle on the output **flips** the answer of the AND; it does not leave it unchanged **[1]** [5]
 Correct version: NAND gives **0 only when both inputs are 1**, and 1 on the other three rows **[1]**
 Evidence: on the row A = 1, B = 1, AND gives 1, so NAND gives 0 **[1]**
 (ii) The mistake: NOT applies only to the thing straight after it, so NOT A AND B means (NOT A) AND B, not NOT (A AND B) **[1]**
 Correct version, shown with a row: with A = 1 and B = 0, (NOT A) AND B gives 0 but NOT (A AND B) gives 1, so they differ **[1]**

Accept any reasonable wording, and give credit for a correct version that is stated without a separate mistake sentence if the correction is explicit. "NAND is the opposite of AND" with no row earns the first two marks of (i) at most. The page's trap list names both of these wrong sentences. Marking them in Rohan's notebook is far easier to face than being told you made them, and it is the same thinking.